

UNIT 1

1.1 Introduction to Web Technology

1.1.1 What is the web?

The web, also known as the World Wide Web (WWW).

The Web (World Wide Web) is a system of websites and webpages that can be accessed through the Internet using browsers.

1.1.2 Evolution of Web

Web 1.0: Static, read-only websites.

Web 2.0: Interactive, user-driven content (social media, blogs).

Web 3.0: AI-driven, decentralized, blockchain-based, semantic web.

1.1.3 Some web

components Web – Page

A document which can be displayed in a web browser such as Firefox, Google Chrome, Opera, Microsoft Internet Explorer or Edge, or Apple's Safari.

Web-Site

Collection of web pages which are grouped together and usually connected together in various ways.

Web Server

A web server is a computer that runs websites.

A web server is a software that Serves web content through the HTTP protocol

1.1.4 Definition of Web Technology

“Web Technology is the technology used to make and run websites on the Internet.”

“Web Technology refers to the collection of tools, languages, and protocols that allow

computers and devices to communicate through the World Wide Web (WWW).”

These technologies allow users to browse websites, interact with online content, and share information over the internet.

1.1.5 Components of Web Technology

1.HTML (Hypertext Markup Language)

- HTML is the foundational markup language of the web.
- It defines the structure of web pages by specifying elements such as headings, paragraphs, lists, images, and links.
- HTML acts as the starting point for any website or web application.

2.CSS (Cascading Style Sheets):

- While HTML defines the structure, CSS is responsible for the appearance.
- It controls the design, including colors, fonts, layout, and responsiveness, ensuring that websites look good across various devices and screen sizes.

3.JavaScript:

- JavaScript is a programming language that adds interactivity to websites.
- It enables dynamic content, such as clickable buttons, animations, and real-time form validation.
- JavaScript is often used alongside HTML and CSS to create seamless user experiences.

4.Web Browsers:

- Browsers like Google Chrome, Mozilla Firefox, and Safari interpret HTML, CSS, and JavaScript and render websites on users' devices.
- They play a crucial role in ensuring a smooth user experience.

5.Web Servers:

- Web servers host websites and serve content over the internet.
- They store files and data that make up a website and deliver them to users when requested.
- Web servers use protocols like **HTTP** (Hypertext Transfer Protocol) and **HTTPS** (HTTP Secure) to communicate with browsers.

6. Web Protocols:

- Protocols are the rules that govern communication between web servers and browsers.
- **HTTP** is the most common protocol, but **HTTPS**, which encrypts data for security, is becoming more prevalent.
- Newer protocols like **HTTP/2** and **HTTP/3** offer faster and more efficient communication.

7. APIs (Application Programming Interfaces):

- APIs are essential for modern web applications.
- They allow different software systems to communicate with each other.
- **Example**, a weather website may use an API to fetch real-time data from a third-party weather service.

8. Databases:

- Databases store and manage data for websites and applications.
- They allow for efficient storage, retrieval, and updating of data.
- Examples of databases used in web technology include **MySQL**, **MongoDB**, and **PostgreSQL**.

1.1.6 Types of Web Technologies

1. Frontend Technologies

Frontend technologies are used to build the user-facing part of a website or web application. They include:

- **HTML, CSS, and JavaScript:** The trio of core languages for designing and developing user interfaces.
- **Frontend Frameworks:** Tools like **React**, **Vue.js**, and **Angular** help developers quickly build complex, interactive user interfaces and manage state.

2. Backend Technologies

Backend technologies are used to build and manage the server-side of web applications. These include:

- **Server-side Languages:** Python, Ruby, PHP, and **Node.js** are popular backend programming languages used to handle business logic, manage databases, and process user requests.
- **Database Management Systems:** **MySQL**, **MongoDB**, and **PostgreSQL** are commonly used to store and retrieve data on the server-side.
- **Web Servers:** **Apache**, **Nginx**, and cloud services like **AWS** and **Google Cloud** host and manage websites.

3. Full-Stack Technologies

Full-stack development involves both frontend and backend development. Frameworks like **Django** (Python), **Ruby on Rails** (Ruby), and **Node.js** (JavaScript) allow developers to handle both client-side and server-side development.

4. Web APIs

Web APIs enable communication between different software components. Some popular API types include:

- **RESTful APIs:** Representational State Transfer (REST) APIs are widely used for client-server communication.
- **GraphQL:** A modern alternative to REST APIs, allowing clients to request only the data they need.
- **WebSockets:** WebSockets allow for real-time, two-way communication between the client and server.

1.2 Web Designing & Development

1.2.1 Web Design

It involves the color scheme, layout, information flow, and all aspects of UI/UX (user interface and user experience).

Web design covers everything related to a website's visual aesthetics and usability

- **Adobe Creative Cloud.** Software like Photoshop and Illustrator help designers create design elements for web pages, such as graphics and logos.
- **Graphic design.** Graphic design involves crafting visuals and layouts to make the website visually appealing and enhance the overall web experience.
- **UI/UX design.** UI/UX design focuses on creating intuitive user interfaces and seamless user experiences by prioritizing ease of use and navigation.
- **Figma.** This browser-based design tool helps designers do collaborative design work and prototyping, as well as create high-fidelity mock-ups.